

ALEXEI OBLOMKOV

Professor of Mathematics

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RESEARCH INTERESTS

Representation theory (double affine Hecke and symplectic reflection algebras, Cherednik algebras); algebraic geometry (Hilbert schemes, Gromov–Witten and Donaldson–Thomas invariants, affine Springer fibers); knot homology and categorification; mathematical physics and integrable systems.

EDUCATION

2005	Ph.D. in Mathematics, Massachusetts Institute of Technology. Advisor: Pavel Etingof.
1998	M.A. in Pure Mathematics, Independent University of Moscow and Moscow State University.

EMPLOYMENT

2020–	Professor, University of Massachusetts, Amherst.
2018 (Spring)	Research Professor, MSRI, Berkeley.
2015–2019	Associate Professor, University of Massachusetts, Amherst.
2009–2015	Assistant Professor, University of Massachusetts, Amherst.
2006–2009	Instructor, then Visiting Assistant Professor, Princeton University.
2005–2006	Postdoctoral Fellow, Institute for Advanced Study.

GRANTS AND AWARDS

2022–2025	NSF DMS-2200798, \$217,038, sole PI.
2018–2021	NSF FRG DMS-1760373, \$300,000 for UMass.
2014–2019	NSF CAREER DMS-1352398, \$420,000, sole PI.
2010–2013	NSF DMS-1001609, \$123,910, sole PI.
2007–2010	NSF DMS-0701367, \$111,300, sole PI.
2018	Simons Fellow in Mathematics.
2014	NSF CAREER Award.
2010–2012	Alfred P. Sloan Research Fellowship.

SELECTED PUBLICATIONS

Selected Preprints and submitted work

5. (with Y. Huang, R. Jiang) *Quot scheme of points on torus knot singularities*, arXiv:2608.16086.
4. (with I. Cavey, E. Gorsky, J. Turner) *Wall-crossing for Hilbert schemes*, arXiv:2607.05624.
3. (with L. Rozansky) *A categorification of representations of $U_q(\mathfrak{gl}_{1|1})$* , arXiv:2510.13039.
2. (with I. Cavey, E. Gorsky, J. Turner) *Newton–Okounkov bodies for nested Hilbert schemes*, arXiv:2510.07420.
1. (with O. Kivinen, G. Wyss) *Discriminants and motivic integration*, arXiv:2503.18449.

Selected Published and accepted papers

10. (with E. Gorsky, O. Kivinen) *The affine Springer fiber – sheaf correspondence*, *Advances in Mathematics* **464** (2025), arXiv:2204.00303.
9. (with L. Rozansky) *Categorical Chern character and braid groups*, *Adv. Math.* **437** (2024), 66 pp., arXiv:1811.03257.
8. (with M. Moreira, A. Okounkov, R. Pandharipande) *Stationary Virasoro constraints for stable pairs on toric varieties*, *Forum Math. Pi* **10** (2022), 62 pp.
7. (with L. Rozansky) *Knot homology and sheaves on the Hilbert scheme of points on the plane*, *Selecta Math. (N.S.)* **24** (2018), no. 3, 2351–2454, arXiv:1608.03227.
6. (with V. Shende, J. Rasmussen) *The Hilbert scheme of a plane curve singularity and HOMFLY homology of its link*, *Geom. Topol.* **22** (2018), no. 2, 645–691, arXiv:1201.2115.
5. (with E. Gorsky, V. Shende, J. Rasmussen) *Torus knots and rational DAHA*, *Duke Math. J.* **163** (2014), no. 15, 325–401, arXiv:1207.4523.
4. (with V. Shende) *The Hilbert scheme of a plane curve singularity and the HOMFLY polynomial of its link*, *Duke Math. J.* **161** (2012), no. 7, 1271–1303, arXiv:1003.1568.
3. (with D. Maulik, A. Okounkov, R. Pandharipande) *The Gromov–Witten/Donaldson–Thomas correspondence for toric 3-folds*, *Invent. Math.* **186** (2011), no. 2, 435–479, arXiv:0809.3976.
2. (with D. Maulik) *Quantum cohomology of $\text{Hilb}_m(A_n)$* , *J. Amer. Math. Soc.* **22** (2009), 1055–1091, arXiv:0802.2737.
1. (with P. Etingof, W. L. Gan, V. Ginzburg) *Harish-Chandra homomorphisms and symplectic reflection algebras for wreath-products*, *Publ. Math. IHÉS* **105** (2007), 91–155, arXiv:math.RT/0511489.

The complete list of 52 papers is available at oblomkov-math.github.io.

EDITORIAL SERVICE

2025– **Managing Editor, *Galileo***
Mathematical Sciences Publishers (MSP); first issue 2027.

Referee for *Annals of Mathematics*, *Inventiones*, *Journal of the AMS*, *Communications in Mathematical Physics*, *Advances in Mathematics*, *Compositio*, *Selecta*, *IMRN*, and *Transactions of the AMS*. Grant proposal review for NSERC, NSA, Swiss NSF, NWO, and the Newton Institute; three NSF panels.

SELECTED CONFERENCE ORGANIZATION

2025 Month-long program, Bernoulli Center, EPFL, Lausanne.
 2023 Two-month program, Simons Center for Geometry and Physics.
 2021–2023 AIM research community and AIM workshop.
 2019–2022 FRG workshops at UC Davis, UMass Amherst and the University of Oregon.
 2014–2018 Workshops at Banff, Amherst and AIM.

PH.D. ADVISING

2024 Arthur Wang.
 2016 Thomas Shelly — Visiting Assistant Professor, Mount Holyoke College.
 2015 Tobias Wilson — Bloomberg.
 Current Pranav Kalkunte; Chuijiao Zhang.

Mentored UMass REU teams in the summers of 2013, 2015–2019 and 2025.